

EXPLAINABLE COVID-19 CHEST X-RAY CLASSIFICATION USING DENSENET121 AND GRAD-CAM

Internship Project Report

Submitted By

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Project Domain

**Computer Vision and Explainable Artificial
Intelligence (XAI)**

Technologies Used

- Python
- TensorFlow / Keras
- DenseNet121
- OpenCV
- Grad-CAM
- Integrated Gradients
- NumPy
- Matplotlib
- Scikit-Learn

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1. Introduction

Deep learning models have demonstrated remarkable performance in medical image classification tasks; however, their black-box nature limits their adoption in clinical environments. This project presents an Explainable Artificial Intelligence (XAI) framework for classifying chest X-ray images into COVID-19, Normal, and Viral Pneumonia categories using DenseNet121. To improve interpretability, Grad-CAM and Integrated Gradients were employed to visualize regions contributing to model decisions. The trained model achieved a training accuracy of 98.30%, validation accuracy of 96.33%, and test accuracy of 97.47%. Quantitative explanation evaluation was performed using Entropy, Deletion AUC, and Area Over the Perturbation Curve (AOPC). Experimental results demonstrate that the model not only achieves high classification performance but also provides meaningful visual explanations that increase transparency and trustworthiness.

Keywords: Deep Learning, Explainable AI, DenseNet121, COVID-19, Chest X-ray, Grad-CAM, Integrated Gradients, Medical Imaging

2. Objectives

The primary objectives of this project are:

1. Develop a robust deep learning model for chest X-ray classification.
2. Classify chest X-rays into:
 - COVID-19
 - Normal
 - Viral Pneumonia
3. Achieve high classification performance using transfer learning.
4. Improve model transparency using Explainable AI techniques.
5. Evaluate explanation quality using quantitative metrics.
6. Analyze both correct and incorrect predictions to understand model behavior.

3. Data Preprocessing

Medical image preprocessing plays a critical role in improving model performance and training stability.

The following preprocessing steps were applied:

3.1 Image Resizing

All images were resized to:

224×224 pixels

This resolution matches the input requirements of DenseNet121.

3.2 Image Normalization

DenseNet121 preprocessing was applied using:

`preprocess_input()`

This operation normalizes pixel values according to ImageNet statistics.

3.3 Data Augmentation

To improve model generalization and reduce overfitting, ImageDataGenerator was used with:

- Rotation
- Horizontal Flip
- Zoom
- Width Shift
- Height Shift

Data augmentation artificially increases dataset diversity and improves robustness.

4. Training Results

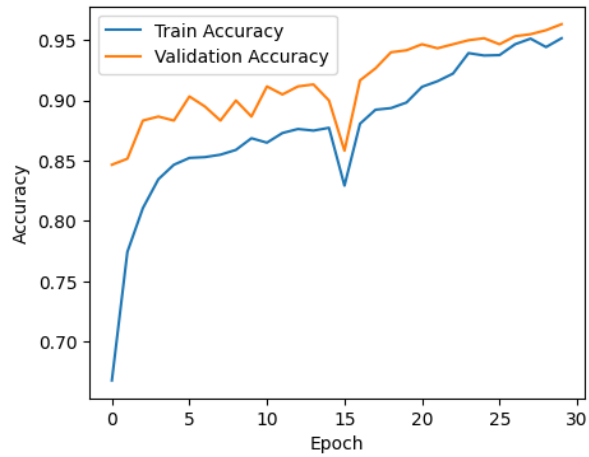
4.1 Accuracy Curves

The training and validation accuracy curves demonstrate consistent learning behavior.

Observations:

- Training accuracy increased steadily.
- Validation accuracy closely followed training accuracy.
- No severe overfitting was observed.

Figure 2: Training and Validation Accuracy Curve



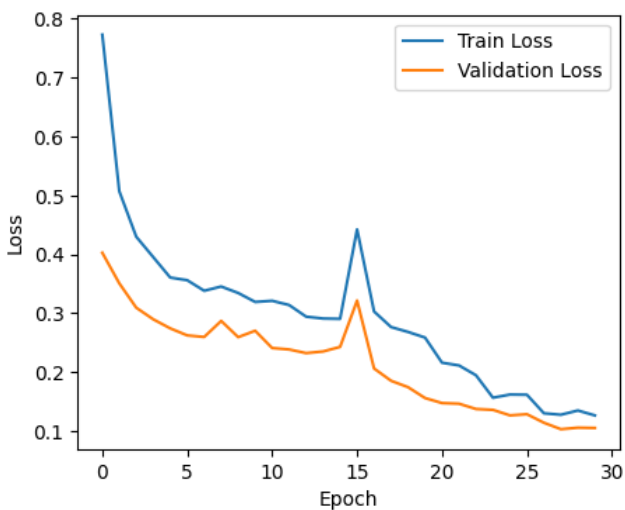
4.2 Loss Curves

Training and validation loss decreased consistently throughout training.

Observations:

- Training loss decreased significantly.
- Validation loss stabilized after convergence.
- Loss trends indicate effective optimization.

Figure 3: Training and Validation Loss Curve



4.3 Classification Report

Overall Accuracy

Metric	Value
Training Accuracy	98.30%
Validation Accuracy	96.33%
Test Accuracy	97.47%

The model achieved strong performance across all dataset splits, indicating good generalization capability.

The Viral Pneumonia class achieved the highest recall, while all classes demonstrated strong precision and F1-scores.

	precision	recall	f1-score	support
COVID	0.98	0.95	0.97	145
Normal	0.95	0.97	0.96	145
Viral Pneumonia	0.99	1.00	1.00	145
accuracy			0.97	435
macro avg	0.97	0.97	0.97	435
weighted avg	0.97	0.97	0.97	435

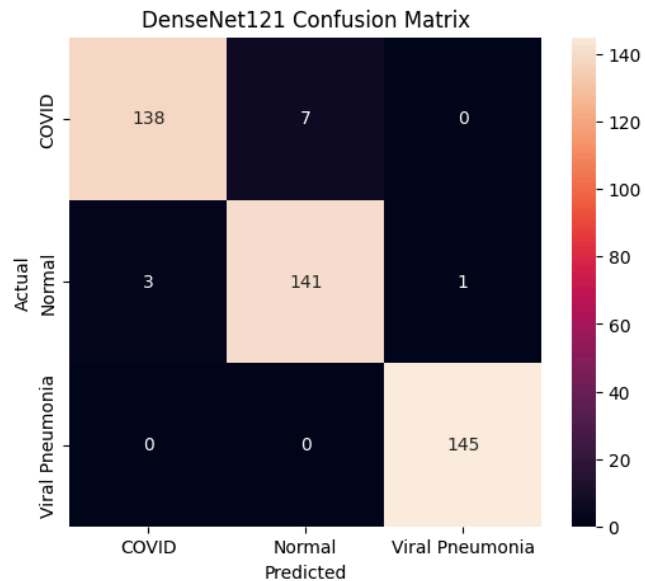
4.4 Confusion Matrix

The confusion matrix provides a detailed view of class-wise prediction performance.

Observations:

- Most samples were correctly classified.
- Only a small number of COVID images were misclassified as Normal.
- Viral Pneumonia showed excellent separability.

Figure 4: Confusion Matrix



5. Explainable Artificial Intelligence (XAI)

Deep learning models often operate as black boxes. Explainability techniques were applied to understand how the model reaches its decisions.

5.1 Grad-CAM

Grad-CAM generates heatmaps highlighting image regions that contribute most to a specific prediction.

Advantages:

- Easy to interpret
- Computationally efficient
- Widely used in medical imaging

5.2 Integrated Gradients

Integrated Gradients is an attribution-based explanation method that quantifies feature importance by integrating gradients along a path from a baseline image to the input image.

Advantages:

- Strong theoretical foundation
- Pixel-level importance estimation

6. Quantitative Evaluation of Explanations

6.1 Entropy Analysis

Entropy measures the concentration of explanation maps. Lower entropy indicates more focused explanations.

Method	Mean Entropy	Standard Deviation
Grad-CAM	10.036	0.39

The obtained entropy values suggest that Grad-CAM focuses on specific image regions rather than distributing attention uniformly.

6.2 Deletion Metric

The Deletion Metric evaluates explanation faithfulness by progressively removing important pixels identified by Grad-CAM and observing confidence degradation.

Results

Metric	Value
Mean Deletion AUC	0.4475
Standard Deviation	0.3784

Lower Deletion AUC values indicate more faithful explanations.

6.3 AOPC Evaluation

Area Over the Perturbation Curve (AOPC) measures the average reduction in confidence after perturbing important image regions.

Results

Metric	Value
AOPC	0.5002

Higher AOPC values indicate stronger explanations of faithfulness.

7. Visual Explanation Analysis

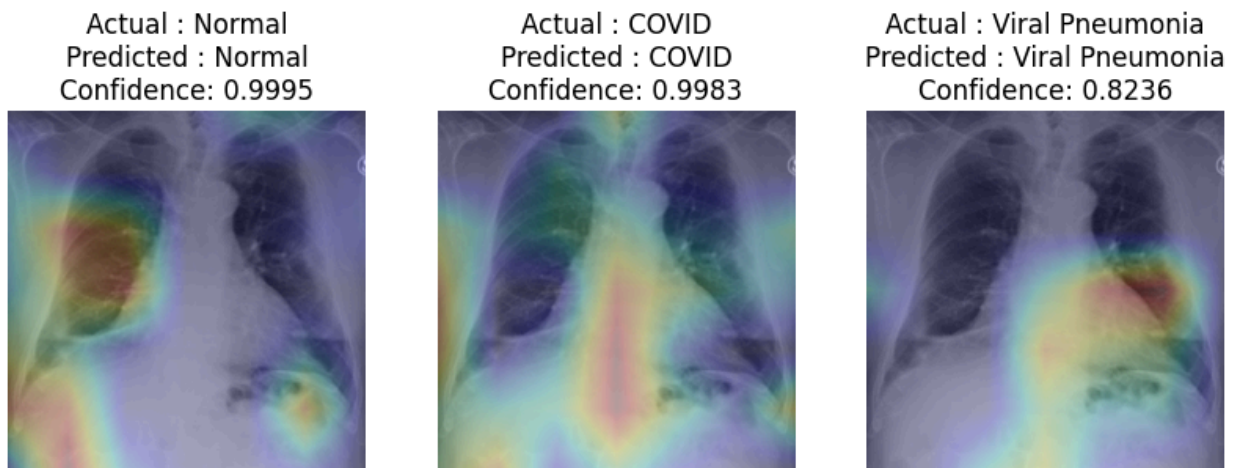
7.1 Correctly Classified Samples

Grad-CAM visualizations were generated for correctly classified images.

Observations:

- COVID predictions focused primarily on pulmonary regions.
- Normal samples showed attention patterns associated with healthy anatomy.
- Viral Pneumonia images exhibited broader activation patterns.

Figure 5: Correctly Classified Samples with Grad-CAM



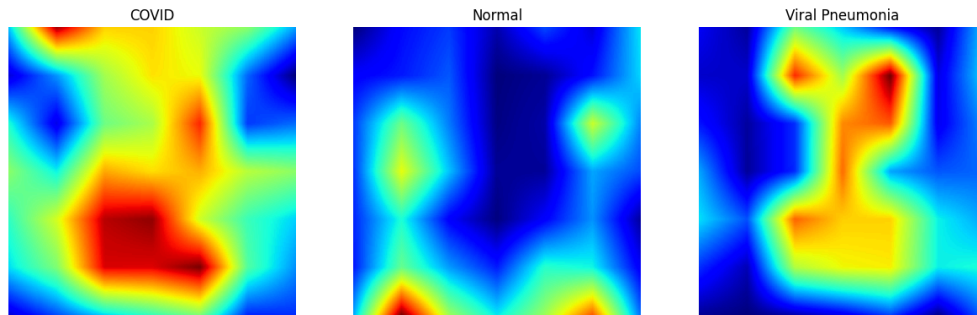
7.2 Average Class-wise Heatmaps

Average Grad-CAM heatmaps were generated for each class.

Findings:

- COVID heatmaps showed concentrated activation in central lung regions.
- Normal heatmaps displayed more distributed attention.
- Viral Pneumonia heatmaps highlighted broader pulmonary areas.

Figure 6: Average Class-wise Heatmaps



7.3 Misclassification Analysis

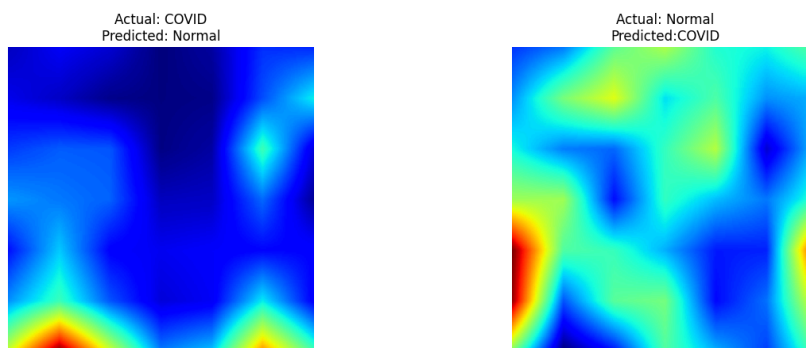
Misclassified samples were examined to understand model failures.

Key observations:

1. Some COVID images exhibited visual patterns similar to Normal cases.
2. Diffuse abnormalities occasionally caused confusion.
3. In certain cases, attention shifted away from disease-specific regions.

Grad-CAM helped identify potential causes of these misclassifications.

Figure 7: Misclassified Samples and Corresponding Grad-CAM Maps



8. Future Work

Future improvements may include:

1. Integration of lung segmentation before classification.
2. Evaluation on larger external datasets.
3. Application of SHAP and LIME for additional explainability.
4. Multi-disease classification beyond COVID-19 and Viral Pneumonia.